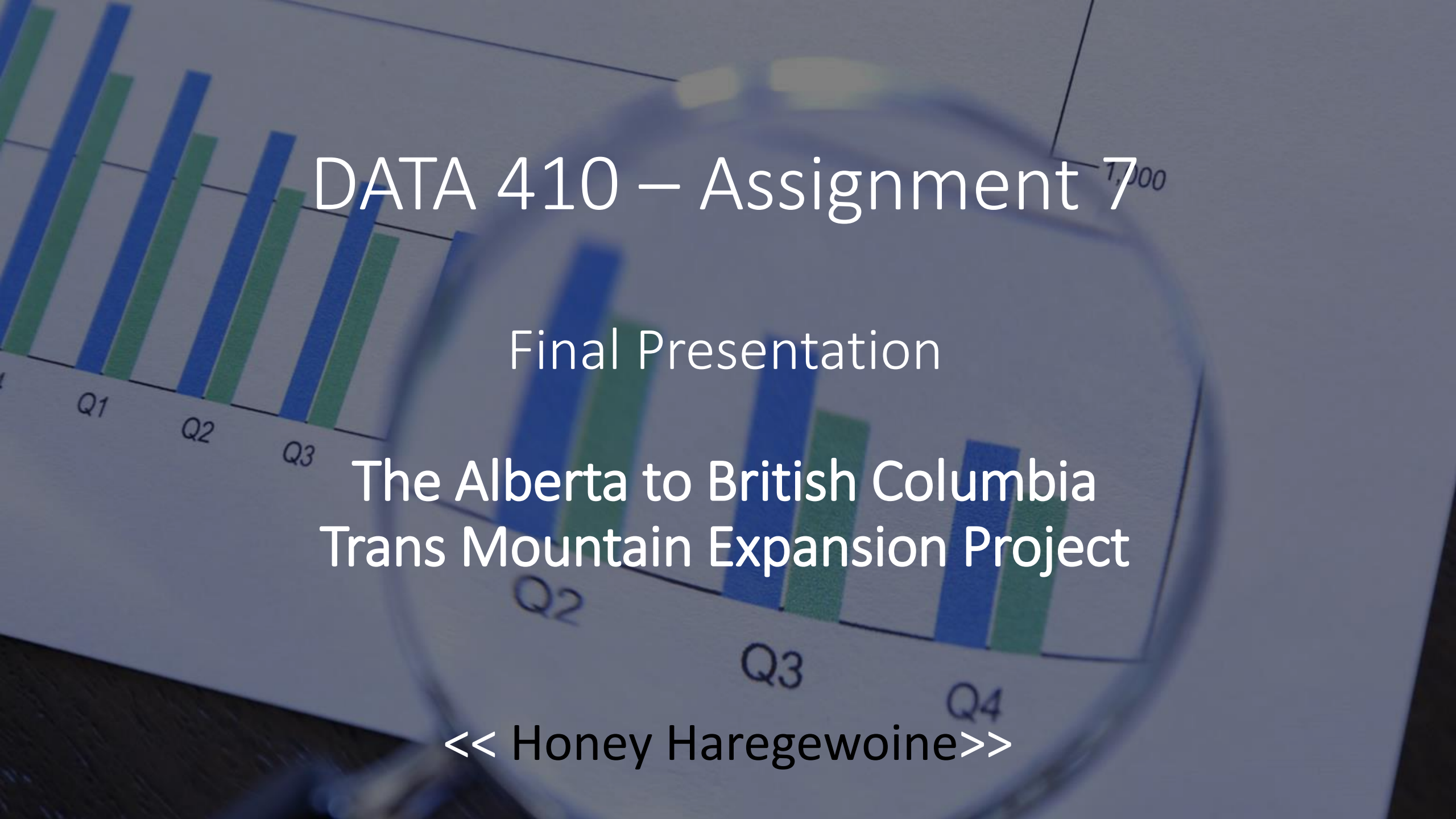




DATA 410 – Assignment 7

Final Presentation

The Alberta to British Columbia
Trans Mountain Expansion Project

<< Honey Haregewoine >>

A magnifying glass is positioned over a bar chart. The chart displays data for four quarters: Q1, Q2, Q3, and Q4. Each quarter has two bars, one blue and one green. The blue bars are consistently taller than the green bars. A horizontal line is drawn across the chart at the 1,000 mark on the y-axis. The magnifying glass is focused on the Q3 and Q4 data points.

1

Business Understanding – High Level

Case Study

- THE ALBERTA TO BRITISH COLUMBIA TRANS MOUNTAIN EXPANSION PROJECT
- The Alberta to British Columbia Trans Mountain Expansion Project

Introduction

- The Trans Mountain Pipeline Expansion Project is a proposal to expand the existing Trans Mountain Pipeline, which transports crude oil from Alberta to British Columbia and Washington state. While the project promises economic benefits, it has sparked significant controversy due to its potential environmental risks, social impacts, and opposition from Indigenous communities. Concerns include the threat of oil spills, harm to marine ecosystems, and violations of Indigenous rights. This discussion explores the various perspectives on the TMEP, highlighting its broader implications for Canada's energy future and Indigenous sovereignty.

Problem Solving Process

1 – Exploring the mess

#	Question	Answer	
Q1.1	What problems (or opportunities) did they face?	Problems • Environmental • Indigenous opposition • Political • Economical	Opportunities - Business expansion/ Economical growth - Job creation - Improved infrastructure - Market opportunities
Q1.2	Where was there a gap between the current situation and the desired one?	The desired situation was to complete the TMEP within the initial cost of \$7.4 billion. However, due to delays and unforeseen challenges, the cost increased to \$12.6 billion.	
Q1.3	What were the stated and unstated goals?	Stated goals • Increase the capacity of the existing pipelines • Enhance Canada's energy	Unstated goals - Strengthen Canada's position in global energy markets - Ensure the long-term

Problem Solving Process

2 – Searching for information

#	Question	Answer										
Q2.1	What were the symptoms and causes?	<table><tr><th><u>Symptoms</u></th><th><u>Causes</u></th></tr><tr><td>Opposition from Indigenous people</td><td>Opposition from Indigenous Groups</td></tr><tr><td>Delays</td><td>Global Market Uncertainty</td></tr><tr><td>Rising project cost</td><td>Cost Underestimation</td></tr><tr><td>Environmental issues</td><td>Regulatory and Legal Challenges</td></tr></table>	<u>Symptoms</u>	<u>Causes</u>	Opposition from Indigenous people	Opposition from Indigenous Groups	Delays	Global Market Uncertainty	Rising project cost	Cost Underestimation	Environmental issues	Regulatory and Legal Challenges
<u>Symptoms</u>	<u>Causes</u>											
Opposition from Indigenous people	Opposition from Indigenous Groups											
Delays	Global Market Uncertainty											
Rising project cost	Cost Underestimation											
Environmental issues	Regulatory and Legal Challenges											
Q2.2	What measures of effectiveness seemed appropriate?	The project's effectiveness can be measured by its ability to stay on schedule, control costs, minimize environmental impact, and address Indigenous concerns, while delivering long-term economic benefits.										
Q2.3	What actions were available?	Available actions included better consultations with Indigenous groups, improving environmental protections, adjusting the budget and timeline, and exploring alternative transportation methods.										

Problem Solving Process

3 – Identifying a problem

#	Question	Answer
Q3.1	Which was the most important problem in this situation?	The most important problem in this situation was balancing economic development with environmental protection and Indigenous rights, as the TMEP faced strong opposition due to its potential environmental risks, legal challenges, and the infringement on Indigenous sovereignty.
Q3.2	Was the problem like others they had dealt with?	The problem was similar to past challenges balancing economic growth with environmental and social responsibility. However, it was more complex because of strong opposition from Indigenous communities, ongoing legal battles, and the risk of long-term environmental damage, making it more controversial than typical infrastructure projects.
Q3.3	What were the consequences of a broad versus narrow problem statement?	A broad problem statement would have led to a general approach, missing specific issues like Indigenous rights or environmental concerns, which could cause more opposition. A narrow problem statement, however, focused on key issues like legal challenges and environmental impact, allowing for more specific

Problem Solving Process

4 – Searching for solutions

#	Question	Answer
Q4.1	What decisions were open to them?	They had several options, such as negotiating with stakeholders for mutually beneficial agreements, improving regulatory compliance, exploring alternative transportation methods to reduce environmental impact, and enhancing communication with Indigenous communities to address their concerns. They could also reconsider the project's scope or delay construction to tackle legal and environmental challenges.
Q4.2	What solutions had been tried in similar situations?	Similar solutions included engaging with stakeholders for compromises, improving environmental assessments, exploring alternative transportation methods, and enhancing consultation to address legal and environmental challenges.
Q4.3	How did the various candidate solutions link to the outcomes of interest?	The candidate solutions, like negotiating with stakeholders and improving regulations, aimed to reduce environmental risks, address legal and social concerns, and ensure the project's success while managing opposition.

Problem Solving Process

5 – Evaluating solutions

#	Question	Answer
Q5.1	How did the solution impact each of the criteria?	The solution positively impacted environmental and social concerns by reducing risks through stricter regulations and stakeholder collaboration. It also addressed legal issues by ensuring better compliance and helped the project move forward while managing opposition. However, it did not fully resolve the financial and timing challenges.
Q5.2	What factors within their control could have improved the outcomes?	Factors within their control that could have improved outcomes include better early engagement with Indigenous communities, more transparent communication with stakeholders, stricter adherence to environmental regulations, and more effective management of project costs and timelines.
Q5.3	What factors outside their control could have altered the outcomes?	Factors outside their control that could have altered the outcomes include changes in government policies, unforeseen legal challenges, global economic conditions affecting oil prices, and unexpected environmental incidents such as oil spills.

Problem Solving Process

6 – Implementing a solution

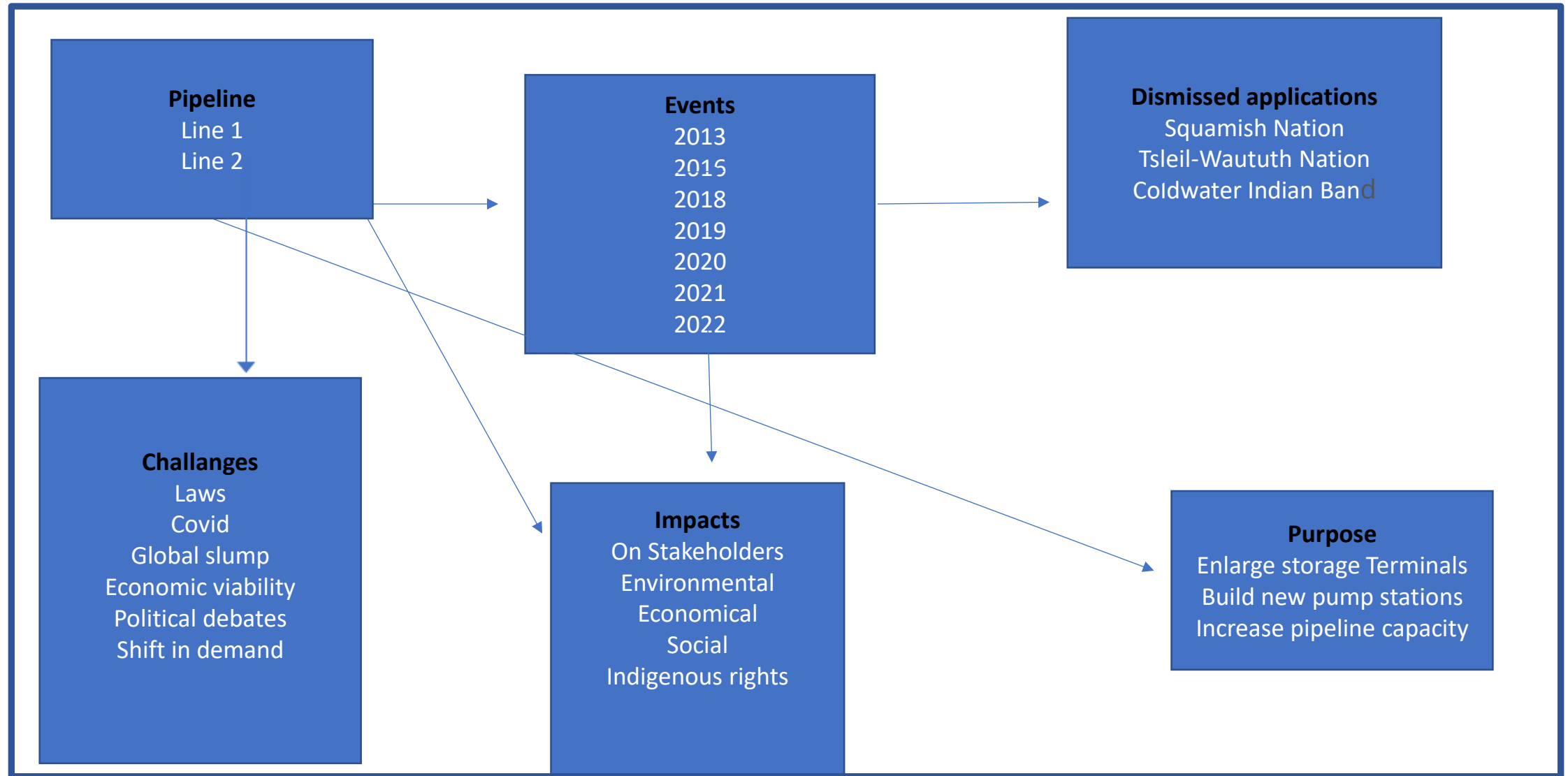
#	Question	Answer
Q6.1	What were the barriers to successful implementation?	Barriers to successful implementation included strong opposition from Indigenous communities, regulatory and legal hurdles, environmental concerns, rising project costs, and public protests. These factors created delays and uncertainty around the project's completion.
Q6.2	Where was there support and motivation, or resistance and conflict?	Support and motivation came from stakeholders seeking economic benefits, such as job creation, energy security, and increased revenue from oil exports. However, resistance and conflict arose primarily from Indigenous communities, environmental groups, and local residents who were concerned about the potential environmental impacts, violations of Indigenous rights, and the long-term social consequences of the project.
Q6.3	Were there resources available for successful implementation?	Yes, there were resources available for successful implementation, such as financial support (even with the cost increase), government approval, technical expertise, and infrastructure. However, issues like regulatory approval,

A magnifying glass is positioned over a bar chart. The chart has a light blue background and features two series of bars: blue and green. The x-axis is labeled with quarters Q1, Q2, Q3, and Q4. The y-axis has a label '1,000' at the top. The magnifying glass is centered over the Q3 and Q4 bars, making them appear larger and more detailed. The text '2 -' is overlaid on the chart, positioned to the left of the Q3 label.

2 -

Data Understanding – High Level

Data Modeling – Conceptual Model



Data Dictionary

No	Topic	Definition
1	TMEP	Trans Mountain Expansion Project
2	NEB	National Energy Board
3	CEO	Chief Executive Officer
4	Pipeline work camps (Man campus)	Large temporary settlements of workers near pipeline construction sites
5	Skwxwú7mesh(Squamish)	Squamish people's exclusive and distinctive language.
6	FPIC	Free, Prior, and Informed Consent.
7	UNDRIP	United Nations Declaration on the Rights of Indigenous Peoples
8	CBD	Convention on Biological Diversity
9	ILO	International Labour Organization Convention
10	NEB	National Energy Board

1 (Revisited) –
Business Understanding –
Low Level

Business Questions

- Question 1) What trade type was the most prevalent in the last years?
- Question 2) In which year was the highest total capacity observed, and what was the total amount?



2 (Revisited) – Data Understanding – Low Level

Data Set(s)

- [trans-mountain-throughput-and-capacity-dataset](#)
- [trans-mountain-throughput-and-capacity-dataset](#)

Database Diagram – Tables and Columns

trans-mountain-throughput-and-capacity-dataset	
	Year
	[Corporate Entity]
	[Pipeline Name]
	[Trade Type]
	[Available Capacity]

A magnifying glass is positioned over a bar chart. The chart displays two data series, one in blue and one in green, across four quarters labeled Q1, Q2, Q3, and Q4. The magnifying glass is centered over the Q2 and Q3 bars, making them appear larger and more detailed. The background of the chart is a light blue grid. The text '3 — Data Preparation' is overlaid on the image in a large, white, sans-serif font.

3 — Data Preparation

Data Preparation – Data cleansing

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	Month	Year	Corporate Entity	Pipeline Name	Key Point	Latitude	Longitude	Direction of	Trade Type	Product	Throughput	Nameplate	Available Capacity (1	Reason For Variance		
2			2006	Trans Mountain	Trans Mountain Pipeline	Burnaby	49.2685	-122.931	west	intracanada	domestic heavy	0		35		
3		1	2006	Trans Mountain	Trans Mountain Pipeline	Burnaby	49.2685	-122.931	west	intracanada	domestic light	8		35		
4		1	2006	Trans Mountain	Trans Mountain Pipeline	Burnaby	49.2685	-122.931	west	intracanada	refined petroleum products	13		35		
5		1	2006	Trans Mountain	Trans Mountain Pipeline	Sumas	49.0766	-122.203	south	export	domestic heavy	1		35		
6		1	2006	Trans Mountain	Trans Mountain Pipeline	Sumas	49.0766	-122.203	south	export	domestic light	15		35		
7		1	2006	Trans Mountain	Trans Mountain Pipeline	Westridge	49.2908	-122.95	west	export	domestic heavy	4		35		
8		1	2006	Trans Mountain	Trans Mountain Pipeline	Westridge	49.2908	-122.95	west	export	domestic light	0		35		
9		2	2006	Trans Mountain	Trans Mountain Pipeline	Burnaby	49.2685	-122.931	west	intracanada	domestic heavy	0		36		
0		2	2006	Trans Mountain	Trans Mountain Pipeline	Burnaby	49.2685	-122.931	west	intracanada	domestic light	8		36		
1		2	2006	Trans Mountain	Trans Mountain Pipeline	Burnaby	49.2685	-122.931	west	intracanada	refined petroleum products	14		36		
2		2	2006	Trans Mountain	Trans Mountain Pipeline	Sumas	49.0766	-122.203	south	export	domestic heavy	2		36		
3		2	2006	Trans Mountain	Trans Mountain Pipeline	Sumas	49.0766	-122.203	south	export	domestic light	12		36		
4		2	2006	Trans Mountain	Trans Mountain Pipeline	Westridge	49.2908	-122.95	west	export	domestic heavy	3		36		
5		2	2006	Trans Mountain	Trans Mountain Pipeline	Westridge	49.2908	-122.95	west	export	domestic light	0		36		



Year	Corporate Entity	Pipeline Name	Trade Type	Available Capacity (1000 m3/d)
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	intracanada	35
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	intracanada	35
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	intracanada	35
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	export	35
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	export	35
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	export	35
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	export	35
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	intracanada	36
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	intracanada	36
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	intracanada	36
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	export	36
2006	Trans Mountain Pipeline ULC	Trans Mountain Pipeline	export	36

Data Preparation – SQL Query for Question 1

---- Q.1 What trade type was the most prevalent in the last years?

```
SELECT YEAR,
       [Trade Type],
       SUM(CASE WHEN [Trade Type] = 'intracanada' THEN 1 ELSE 0 END)
  AS Intracanada_Count,

       SUM(CASE WHEN [Trade Type] = 'export' THEN 1 ELSE 0 END)
  AS Export_Count

FROM dbo.[trans-mountain-throughput-and-capacity-dataset]

WHERE YEAR = (SELECT MAX(YEAR)
              FROM dbo.[trans-mountain-throughput-and-capacity-dataset])

GROUP BY YEAR, [Trade Type];
```

Data Preparation – Data set returned for Query 1

YEAR	Trade Type	Intracanada_Count	Export_Count
2020	export	0	24
2020	intracanada	18	0

```
4 SELECT YEAR,  
5     [Trade Type],  
6     SUM(CASE WHEN [Trade Type] = 'intracanada' THEN 1 ELSE 0 END)  
7 AS Intracanada_Count,  
8  
9     SUM(CASE WHEN [Trade Type] = 'export' THEN 1 ELSE 0 END)  
10 AS Export_Count  
11  
12 FROM dbo.[trans-mountain-throughput-and-capacity-dataset]  
13  
14 WHERE YEAR = (SELECT MAX(YEAR)  
15               FROM dbo.[trans-mountain-throughput-and-capacity-dataset])  
16  
17 GROUP BY YEAR, [Trade Type];  
18
```

100 %

Results Messages

	YEAR	Trade Type	Intracanada_Count	Export_Count
1	2020	export	0	24
2	2020	intracanada	18	0

Data Preparation – SQL Query for Question 2

---- Q.2 In which year was the highest total capacity observed, and what was the total amount

```
SELECT Year, SUM([Available Capacity ]) AS Total_Capacity  
FROM dbo.[trans-mountain-throughput-and-capacity-dataset]
```

```
GROUP BY Year  
ORDER BY Total_Capacity DESC
```


Data Preparation – Data set returned for Query 2

	Year	Total_Capacity
1	2019	4214
2	2016	4179
3	2017	4137
4	2015	4123
5	2010	3990
6	2018	3857
7	2014	3780
8	2012	3752
9	2009	3689
10	2011	3654
11	2013	3612
12	2008	3381
13	2007	3353
14	2006	2961
15	2020	2016

```
1 -----
2 ---- Q.2 In which year was the highest total capacity observed, and what was the total amount
3
4 SELECT Year, SUM([Available Capacity ]) AS Total_Capacity
5 FROM dbo.[trans-mountain-throughput-and-capacity-dataset]
6
7 GROUP BY Year
8 ORDER BY Total_Capacity DESC
```

100 %

Results Messages

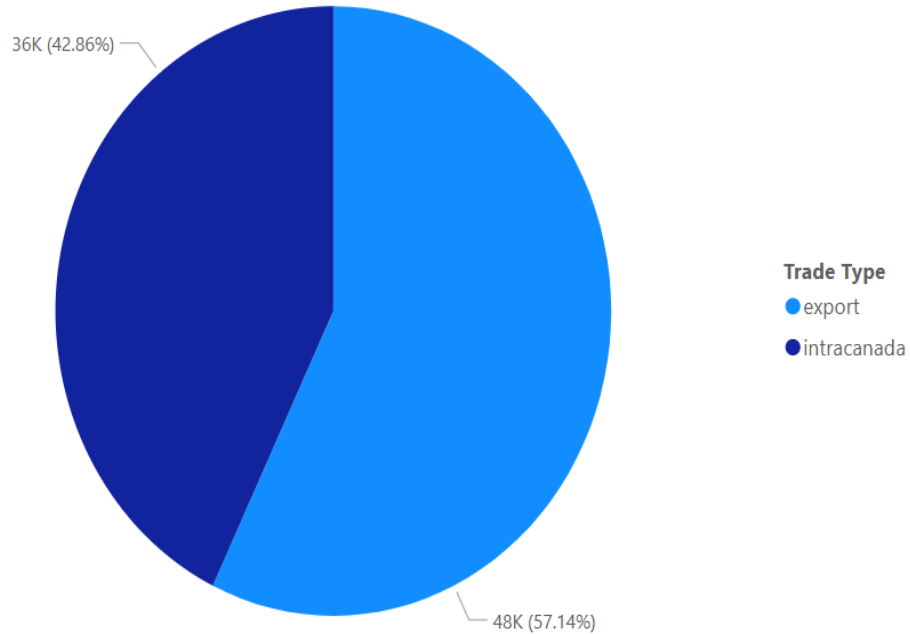
	Year	Total_Capacity
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15	2020	2016

A magnifying glass is positioned over a bar chart, focusing on the Q3 and Q4 data points. The chart features blue and green bars for each quarter. A horizontal line at the 1,000 mark is visible on the right side of the chart. The text '4 - Modeling' is overlaid on the left side of the image.

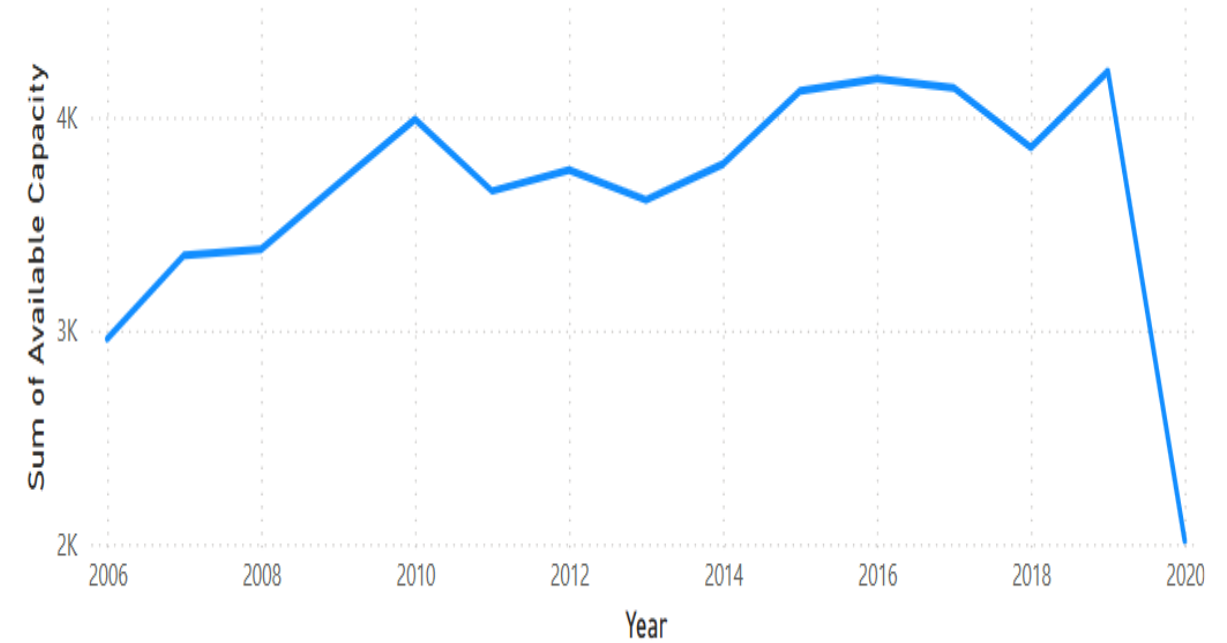
4 - Modeling

Visualization - Dashboard

Sum of Year by Trade Type



Sum of Available Capacity by Year



Visualization – Report 1

Trade Type	Sum of Year
export	48480
intracanada	36360
Total	84840

Visualization – Report 2

Year	Sum of Available Capacity
2019	4214
2016	4179
2017	4137
2015	4123
2010	3990
2018	3857
2014	3780
2012	3752
2009	3689
2011	3654
2013	3612
2008	3381
2007	3353
2006	2961
2020	2016
Total	54698

5 – Evaluation



Conclusion – Answer to Question 1

- In 2020, a total of 42 trades were recorded. Of these, 24 were exports, representing a significant portion of the total trade activities. This indicates that the export trade type was the most common during that year, accounting for approximately 57% of all trades. The remaining 18 trades were classified as intracanada, showing that exports played a more prominent role in 2020 compared to intracanada trade type.

Conclusion – Answer to Question 2



- Between 2006 and 2020, the year that recorded the highest total capacity was 2019, with a total capacity of 4,214. This figure represents the peak level of capacity during the given period, highlighting 2019 as the year with the most significant capacity recorded compared to other years in the dataset.